Cambridge (CIE) A Level Maths: Pure 1

Your notes

Geometric Progressions

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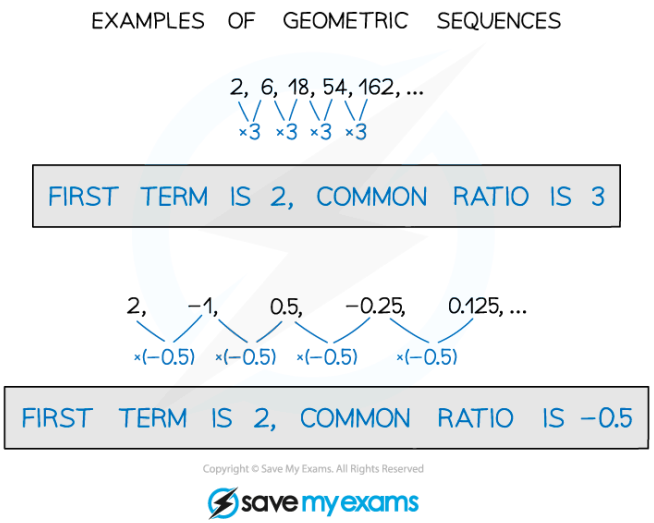
Geometric progressions

What is a geometric progression?

- In a **geometric progression** (also called **geometric sequence**) there is a **common ratio** between consecutive terms in the sequence



- You need to know the ***n*th term formula** for a geometric progression

$$u_n = a r^{n-1}$$

  - *a* is the first term
  - *r* is the common ratio

Examiner Tips and Tricks

- The formula is given in the formula booklet
- If you know two terms in a geometric sequence you can find ***a*** and ***r*** using simultaneous equations.

e.g. THE 6th TERM OF A GEOMETRIC SEQUENCE IS 486.
THE 10th TERM IS 39 366.
FIND THE FIRST TERM AND THE COMMON DIFFERENCE.

$u_n = ar^{n-1}$

$ar^5 = 486$ ①

$ar^9 = 39\,366$ ②

SOLVE THESE SIMULTANEOUS EQUATIONS

$r^4 = \frac{39\,366}{486} = 81$

② DIVIDED BY ①

$r = 81^{\frac{1}{4}} = 3$

$a(3)^5 = 486$

$243a = 486$

$a = 2$

SUBSTITUTE INTO ①

THE FIRST TERM IS 2 AND
THE COMMON RATIO IS 3

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Worked Example

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The first three terms in a geometric sequence are $2 - x$, $3x$, and x^2 respectively where $x < 0$.

a) Show that $x^3 + 7x^2 = 0$.

b) Find the 6th term of the sequence.

a) THIS IS A GEOMETRIC SEQUENCE SO THERE'S A SINGLE COMMON RATIO

$r = \frac{3x}{2-x}$ ← 2nd TERM DIVIDED BY 1st TERM

AND

$r = \frac{x^2}{3x}$ ← 3rd TERM DIVIDED BY 2nd TERM

SO $\frac{3x}{2-x} = \frac{x^2}{3x}$

$(3x)^2 = x^2(2-x)$

$9x^2 = 2x^2 - x^3$

$x^3 + 7x^2 = 0$

b) $x^2(x+7) = 0$ ← SOLVE THE EQUATION FROM a)

$x = 0$ OR $x = -7$

BUT $x < 0$ SO $x = -7$

$a = 2 - (-7) = 9$ ← FIND a AND r

$r = \frac{3(-7)}{2-(-7)} = \frac{-21}{9} = \frac{-7}{3}$

$u_6 = 9\left(\frac{-7}{3}\right)^5 = -\frac{16\,807}{27}$ ← $u_n = ar^{n-1}$

THE SIXTH TERM IS $-\frac{16\,807}{27}$

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Sum of geometric progressions

How do I find the sum of a geometric progression?

- The **sum** of the terms of a **geometric progression** is sometimes called a **geometric series**

1, 3, 9, 27, 81, ... IS A GEOMETRIC SEQUENCE

1 + 3 + 9 + 27 + 81 + ... IS A GEOMETRIC SERIES

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- The following formulae will let you find the sum of the first ***n*** terms of a geometric progression:

$$S_n = \frac{a(1-r^n)}{1-r} \quad \text{or} \quad S_n = \frac{a(r^n-1)}{r-1}$$

  - *a* is the first term
  - *r* is the common ratio
- The one on the left is more convenient if ***r* < 1**, the one on the right is more convenient if ***r* > 1**
- The ***a*** and the ***r*** in those formulae are exactly the same as the ones used with **geometric progression**

How do I prove the formula for the sum of a geometric progression?

- Learn this proof of the sum of a geometric progression formula – you can be asked to give it in the exam:
  - Write out the sum once
  - Write out the sum again but multiply each term by *r*
  - Subtract the second sum from the first
    - All the terms except the two should cancel out
  - Factorise and rearrange to make *S* the subject

THE SUM OF THE FIRST n TERMS IS

$$S_n = a + ar + ar^2 + ar^3 + \ldots + ar^{n-2} + ar^{n-1} \qquad (1)$$

MULTIPLYING BY r GIVES

$$rS_n = ar + ar^2 + ar^3 + ar^4 + \ldots + ar^{n-1} + ar^n \qquad (2)$$

SUBTRACT (2) FROM (1) AND REARRANGE

$$S_n - rS_n = a - ar^n$$

$$S_n(1-r) = a(1-r^n)$$

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

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What is the sum to infinity of a geometric progression?

- If (and only if!) **|r| < 1**, then the sum of a geometric progression **converges** to a finite value given by the formula

$$S_\infty = \frac{a}{1-r}$$

- S_∞ is known as the **sum to infinity**
- If **|r| ≥ 1** the sum of a geometric progression is **divergent** and the sum to infinity does not exist

Worked Example

Your notes

A geometric progression has first term a.

Each term is multiplied by 0.9 to get the next term.

The sum of the first 5 terms is 409.51.

a) Find the value of a.

b) What is the sum to infinity of the progression?

a) IDENTIFY KEY INFORMATION

$a = a$

$r = 0.9$

$n = 5$

$S_5 = 409.51$

SUBSTITUTE INTO THE FORMULA $S_n = \frac{a(1-r^n)}{1-r}$

$409.51 = \frac{a(1-0.9^5)}{1-0.9}$

EVALUATE USING CALCULATOR

$409.51 = 4.0951a$

$a = \frac{409.51}{4.0951}$

$a = 100$

b) USE FORMULA $S_\infty = \frac{a}{1-r}$

$S_\infty = \frac{100}{1-0.9}$

$S_\infty = 1000$

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Examiner Tips and Tricks

The geometric series formulae are in the formulae booklet – you don't need to memorise them.